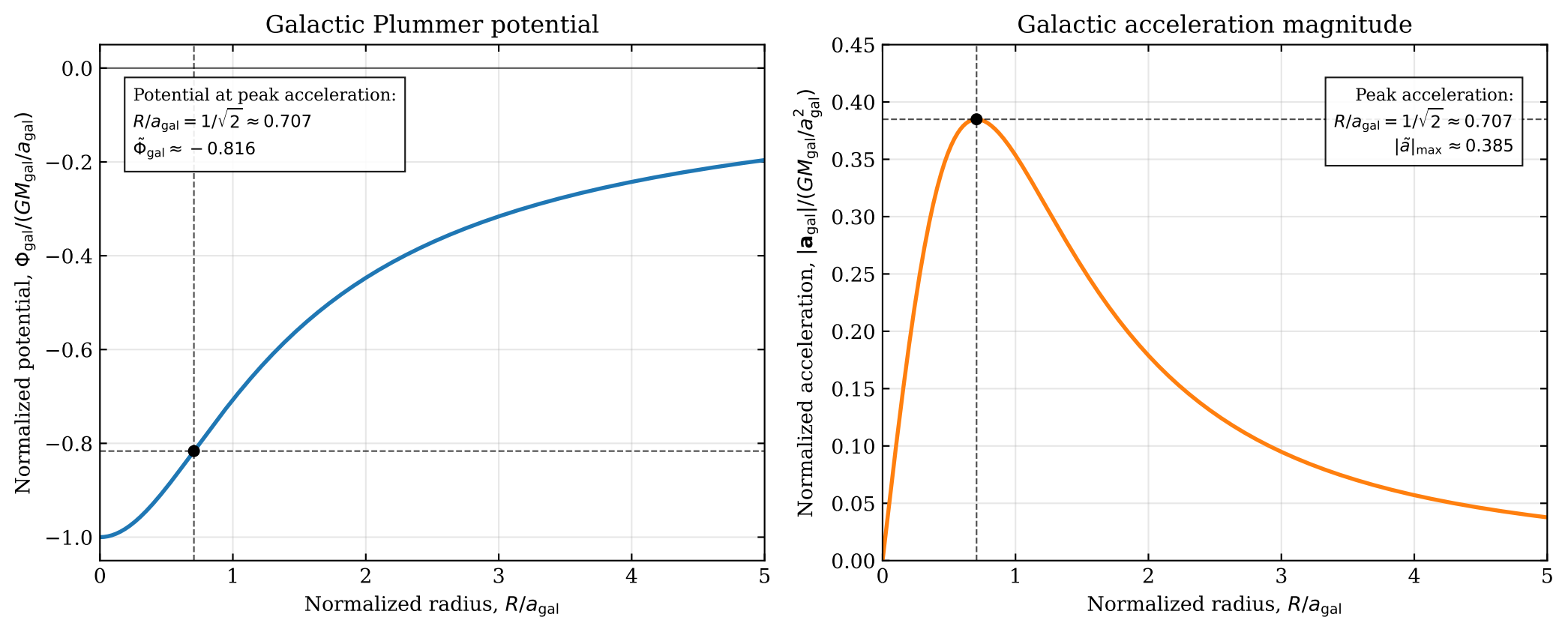
Galactic Plummer potential
Potential at peak acceleration:
$R/a_{\rm gal} = 1/\sqrt{2} \approx 0.707$
$\tilde{\Phi}_{\rm gal} \approx -0.816$
Normalized potential, $\Phi_{\rm gal}/(GM_{\rm gal}/a_{\rm gal})$
Normalized radius, $R/a_{\rm gal}$
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
1
2
3
4
5
Galactic acceleration magnitude
Peak acceleration:
$R/a_{\rm gal} = 1/\sqrt{2} \approx 0.707$
$|\tilde{a}|_{\rm max} \approx 0.385$
Normalized acceleration, $|\mathbf{a}_{\rm gal}|/(GM_{\rm gal}/a_{\rm gal}^2)$
Normalized radius, $R/a_{\rm gal}$
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
1
2
3
4
5